



Christina Ertural

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Nationality: German

in christina-ertural

Living in Berlin

R^G Christina-Ertural

QuantumChemist

PROFILE

Senior Software Architect & Computational Scientist

Results-driven Software Architect, Computational Scientist and Full Stack Quantum Chemist operating at the intersection of High-Performance Computing (HPC), Machine Learning, and complex domain sciences. Proven track record of translating intricate mathematical, chemical, and physical challenges into highly optimized, scalable production code (LOBSTER, AutoPLEX, LoopingPilot). Expert in designing robust data pipelines (ETL) and modular frameworks in Python and C++ with thousands of active users worldwide. Autonomously scales deep-tech infrastructures, pioneers modern software architecture stacks, and aligns technical implementations directly with product and client requirements.

FURTHER TRAINING

06/2026 – 01/2027 Further training for Data Expert
StackFuel GmbH

- BI Specialist in SQL & Relational Databases:
Creation and administration of data bases, data-driven business decisions
- Data Analyst & Data Scientist:
Data analysis (EDA, SQL), predictions, statistics, machine learning and AI-supported decisions
- Project management:
Business Cases, planing resources, Kanban and Jira, management of risks, stakeholders and project quality

PROFESSIONAL EXPERIENCE

05/2026 – Present Co-Founder — AI, Algorithms & Scientific Modeling (Deep-Tech Project in the Pre-Founding Stage for Sustainable Aquaponics)
enkiponics, Berlin (Remote)

- Developing custom AI/ML core models in Python to process multi-sensor data streams (temperature, pH, nutrients, photos) for biological system stabilization.
- Architecting predictive time-series algorithms based on LSTMs, Bayesian Networks, CNNs and YOLO to simulate and prevent ecosystem imbalances in closed-loop agricultural systems.
- Designing a scalable data pipeline integrating real-time sensor IoT inputs with localized database storage (MongoDB/SQL/client-dependent).
- Leading B2B customer discovery interviews to validate market fit, translate client pain points into technical product requirements, and drive cross-functional alignment within the founding team.

09/2022 – 08/2025 Software Architect ML & Materials Informatics (Postdoctoral Researcher)
BAM Federal Institute for Materials Research and Testing, Berlin
(Governmental Research Institute)

- Designed and developed the Python workflow software AutoPLEX to automate large-scale data analysis and machine learning (ML) model generation for material simulations, within the Materials Project software frameworks and based on MongoDB database management, resulting in scientific publications, 150+ stars on GitHub, 20k+ downloads on pepy.tech and a large user base.
- Collaborated in interdisciplinary teams, translating scientific needs into robust code.
- Applied modern open source software engineering practices (version control Git/GitHub, testing, documentation); working on tools like atomate2, pymatgen, jobflow-remote and qtoolkit, which have a huge user base.

12/2017 – 07/2022 Software Developer / Doctoral thesis (Dr. rer. nat.)
RWTH Aachen University (Part of Germany's Excellence Initiative)

- C++ development and implementation of several quantum chemical bonding descriptors for the LOBSTER software (1.5k+ citations and huge user base) for electronic structure analysis, version control Apache Subversion SVN.
- Conducted computational research using VASP, Quantum ESPRESSO, ABINIT, resulting in publications with 2.5k+ citations.
- Supervised and trained students in quantum chemistry, coding and data analysis.

EDUCATION

12/2017 – 07/2022 Doctorate (Dr. rer. nat.), Theoretical Solid State Chemistry
RWTH Aachen University

10/2015 – 09/2017 Master of Science, Chemistry
RWTH Aachen University

10/2011 – 09/2015 Bachelor of Science, Chemistry
RWTH Aachen University

TECHNICAL SKILLS

Programming C++, Python, JavaScript/TypeScript, HTML/CSS, Git, GitHub, Linux, Bash scripting

Frameworks, Tools & More MongoDB, MySQL, HPC environments, Agile collaboration, Open Source Development, Machine Learning, Workshop Organization, Mentoring

LANGUAGES

German (native), English (fluent), French (intermediate)